

LR²: A Legal RAG for Legal Reasoning over Cases

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Abstract. This paper presents a Retrieval Augmented Generation (RAG) pipeline adapted to simulate the process of resolving legal cases. It introduces two novel approaches to enhance the RAG pipeline for legal reasoning: 1) *problem generation*, which automatically creates fictitious legal cases to improve retrieval performance, and 2) *query rewriting*, which reformulates user queries to better align with relevant passages. These methods are evaluated on a legal reasoning dataset derived from real U.S. civil procedure cases. Experiments conducted across different retrieval methods (keyword search, tensor search, and hybrid search with re-ranking) and language models (GPT-4, SAULM, Llama-3 8B) highlight the benefits of the more advanced RAG pipeline. Notably, the query rewriting approach consistently improves performance by enhancing the alignment between queries and passages.

Keywords. Retrieval Augmented Generation, Legal Reasoning

1. Introduction

Artificial Intelligence has significant potential to assist legal professionals by reducing the time and effort required for complex legal tasks. Recent research has explored LLMs' capabilities in legal reasoning through prompting-based approaches for tasks such as legal judgment prediction [1], statutory reasoning [2], and legal exams [3], though limited attention has been given to open-source models [4] or their integration with closed-source systems like GPT-4 [5]. RAG systems have evolved from basic retrieve-read pipelines [6] to advanced modular architectures with adaptive retrieval flows [7–9], though these sophisticated approaches often require substantial computational resources and data [10, 11]. In the legal domain specifically, RAG-based systems have been developed for statutory law inquiries in different languages [12, 13], highlighting the need for more comprehensive approaches to legal question-answering systems.

This paper addresses the task of legal reasoning over court cases, which involves a two-step process: (1) retrieving the most pertinent passages from case text, such as statutory provisions, precedents, or doctrinal explications, and (2) analyzing the retrieved materials to determine the most appropriate resolution. It introduces LR² (Legal RAG

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for Legal Reasoning), a RAG pipeline specifically designed to address query-passage misalignment issues that negatively impact retrieval performance in legal contexts.

2. Methodology

The proposed LR² pipeline builds upon the standard Retrieve-Read RAG architecture [6] with two novel enhancements to address query-passage misalignment:

Problem Generation: Instead of indexing original passages directly, the system automatically generates fictitious legal cases from each original passage using an LLM. For each passage p_j , a corresponding fictitious legal problem p'_j is generated. The retriever then computes similarity scores $s(q_i, p'_j)$ between the user query and generated problems, ranking passages and returning the top-k original passages p_j associated with the top-scores generated problems. This approach aims to better align the knowledge base content with typical user queries.

Query Rewriting: An LLM reformulates the user input q_i to extract the most important terms, producing a refined query q'_i . The retrieval module then computes scores $s(q'_i, p_j)$ for each original passage, while the generation component uses the original user input q_i with the retrieved passages to produce the final answer. This method enhances retrieval effectiveness by improving query-passage alignment.

The dataset used for evaluation is derived from U.S. civil procedure domain [14, 15], converted into a multiple-choice question answering format using Multiple-Choice Prompting (MCP) [16]. Each instance consists of a legal problem question, context introduction, possible solution arguments, and annotated labels indicating correctness.

3. Results

GPT-4.1 demonstrated the best overall performance across all methods, with query rewriting achieving the highest scores (Accuracy: 0.632, Precision: 0.620, Recall: 0.611, F1: 0.613). For GPT-4, query rewriting yielded substantial improvements (Accuracy: 0.590, F1: 0.550) compared to the baseline Retrieve-Read approach (Accuracy: 0.515, F1: 0.414). Llama-3 8B showed modest improvements with query rewriting (Accuracy: 0.410, F1: 0.402) over baseline (Accuracy: 0.401, F1: 0.382). SAULM, despite being pre-trained on legal domain, demonstrated lower generalization capability in zero-shot reasoning tasks. The hybrid BM25 with tensor re-ranking consistently outperformed other methods. Query rewriting with this hybrid approach achieved the best retrieval metrics (MRR: 0.368, R@1: 0.234, R@5: 0.563, R@10: 0.688), outperforming the baseline Retrieve-Read method (MRR: 0.337, R@1: 0.219, R@10: 0.633). Problem generation yielded modest improvements (MRR: 0.359, R@1: 0.211) but did not match query rewriting's effectiveness. Methods relying solely on dense vector representations consistently underperformed compared to keyword-based and hybrid approaches, suggesting that explicit keyword matching remains crucial for legal information retrieval.

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